Municipal Waste Collection Optimiser

Problem Solving Using Programming (Java) | PBL Project | Y26 Section 6, KL University Hyderabad

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Abstract

Municipal waste collection is often done on a fixed schedule. Trucks visit every area every day, even when some bins are almost empty while others are overflowing. This wastes fuel, time and money, and full bins left unattended cause hygiene problems in the city.

The Municipal Waste Collection Optimiser is a console-based Java application that helps a municipality record and analyse the waste produced in different areas. The user enters the waste (in kg) collected from each area for each day of the week. The program stores this data in a two-dimensional array (areas × days) and uses it to support simple, data-based decisions.

The system calculates the fill percentage of each area's bins and classifies them as LOW, HALF, FULL or OVERFLOW. It generates a weekly report showing each area's total and average waste, the city total, the collection cost and the busiest day. Areas are ranked from highest to lowest waste using bubble sort, and any area can be searched by name. For a selected day, the system prepares a collection plan that skips areas with low bins and calculates the number of trucks needed and the estimated cost.

The project applies the concepts learnt in the course: variables, operators, type casting and Scanner input (CO1); if-else ladders, switch-based menus and loops with break and continue (CO2); and methods, method overloading, recursion, and one- and two-dimensional arrays with searching and sorting (CO3).